

# An Introduction to Climate Modeling

## Milestone 3 - Solving a zero-dimensional Energy Balance Model

### 1 Solving a zero-dimensional Energy Balance Model

In the previous milestones you have already implemented some routines that are part of an energy balance model. Now you have the opportunity to combine your results into a zero-dimensional (0D) energy balance model (EBM) and solve the model using an ODE solver. Note that for zero-dimensional EBMs, spatial dependence is neglected, and  $T$  is a global time-dependent averaged surface temperature. Let the following equation be given for the energy balance model:

$$\overline{C} \frac{\partial T_A(t)}{\partial t} + A + BT_A(t) = \overline{S_{\text{sol}}}(t), \quad (1)$$

where  $\overline{C}$  is the mean heat capacity,  $T_A$  is an approximation to the area-average of the time dependent temperature,  $A$  the  $CO_2$  dependent radiative cooling,  $B = 2.15$  the feedback effects such as water vapor cycles, lapse rate and cloud cover,  $\overline{S_{\text{sol}}}(t)$  is the area-averaged solar forcing. To solve the zero-dimensional EBM you should proceed as described in the next steps.

1. Implement a function `calc_area`, which determines the area fraction of each grid cell as a function of latitude, with an equidistant step size  $h = \frac{\pi}{64}$  for a  $128 \times 65$  grid, and returns the values as a vector.
2. Implement a `calc_mean` function, using the results from the first task, which takes a matrix of values of a given parameter like albedo or solar forcing for each grid cell and returns the mean value over the whole area.
3. Write a function `calc_radiative_cooling_co2`, which returns the  $CO_2$  dependent radiative cooling  $A$  for a given  $CO_2$  concentration  $c$ . As default value use the  $CO_2$  concentration of the year 1950:  $c_0 = 315.0$ . The radiative cooling  $A$  is given by

$$A(c) = 210.3 - 5.35 \log \left( \frac{c}{c_0} \right). \quad (2)$$

4. In a function `compute_equilibrium`, solve the above equation (1) using the forward Euler method and calculate an average temperature per year until equilibrium is reached.
5. Write a new function or adapt your existing one to add an option to solve the equation with the backward Euler method instead. Plot your results.

### 2 Bonus

1. Calculate the temperature evolution with a function `calc_evolution` similar to the function in task 4. This time for the parameter  $A$  use transient  $CO_2$  concentrations that you can take from the input file `co2_nasa.dat` containing NASA  $CO_2$ -Data from 1958 to 2025 with measurements taken from the Mauna Loa Observatory in Hawaii<sup>1</sup>. Each row of this  $CO_2$  data lists:

year, month, decimal date, monthly average, seasonal average, # measurements, std. deviation, uncertainty

For the last three entries values of  $-1$ ,  $-9.99$ ,  $-0.99$  mean "missing data" and should be ignored.

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<sup>1</sup><https://gml.noaa.gov/ccgg/trends/mlo.html>

This means that you need to solve the following equation with the backward Euler method:

$$\overline{C} \frac{\partial T_A(t)}{\partial t} + A(t) + BT_A(t) = \overline{S_{\text{sol}}}(t), \quad (3)$$

with

$$A(t) = 210.3 - 5.35 \log \left( \frac{c}{\bar{c}_0(t)} \right), \quad (4)$$

where  $\bar{c}_0(t)$  is the average  $CO_2$  concentration for a given year.

2. Use this transient  $CO_2$  data to compute the average temperature for the year 2020.
3. Create a function `calc_transient_0DEBM` that computes and saves the average temperature and annual temperatures for each year for which the transient  $CO_2$  data is available.

### 3 Control Solutions

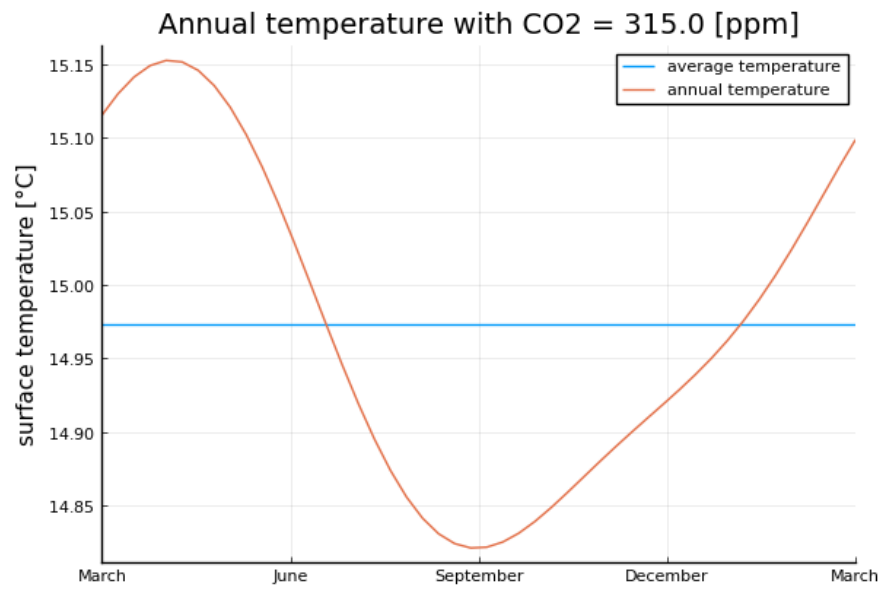


Figure 1: EBM results calculated with forward Euler method (results calculated with backward Euler method are similar)